# 1️⃣ Conditional Statements in Python

Conditional statements allow your program to **make decisions** - run different parts of code based on certain conditions.

**What Are Conditions?**

A **condition** is simply a statement that can be either **True** or **False**.

Example:

```
age = 18
print(age >= 18)   # True
```

**if Statement**

Used to run a block of code only when the condition is **True**.

```
age = int(input("Enter your age: "))

if age >= 18:
    print("You are eligible to vote.")
```

If the condition is false, nothing happens.

**if-else Statement**

```
marks = int(input("Enter your marks: "))

if marks >= 40:
    print("You passed!")
else:
    print("You failed!")
```

**if-elif-else Statement**

Used when we have multiple conditions.

```
marks = int(input("Enter marks: "))

if marks >= 90:
    print("Grade A")
elif marks >= 80:
    print("Grade B")
elif marks >= 70:
    print("Grade C")
else:
    print("Grade D")
```

**Practice Question 1**

Write a Python program that takes a number as input and prints:

- "Positive" if number > 0
- "Zero" if number == 0
- "Negative" if number < 0

## 2️⃣ Lists in Python

**Definition**

A **list** is a built-in data type that can store **multiple values** in a single variable. Lists are **mutable** (can be changed) and can store **different data types**.

**Example:**

```
marks = [87, 64, 33, 95, 76]
foods = ["Samosa", "Pizza", "Burger"]
student = ["Saumya Singh", 21, "Delhi"]
```

**Accessing Elements (Indexing)**

Each item in a list has an index starting from **0**.

```
foods = ["Samosa", "Pizza", "Burger"]
```

```
print(foods[0])    # Samosa
print(foods[2])    # Burger
```

### Modifying Elements

Lists are **changeable**.

```
foods[0] = "GulabJamun"
print(foods)   # ['GulabJamun', 'Pizza', 'Burger']
```

### List Slicing

You can extract parts of a list using slicing.

```
marks = [87, 64, 33, 95, 76]
print(marks[1:4])    # [64, 33, 95]
print(marks[:3])     # [87, 64, 33]
print(marks[-3:-1])  # [33, 95]
```

### List Functions

| Function | Description | Example |
|---|---|---|
| `len(list)` | Returns length of list | `len(marks)` → 5 |
| `max(list)` | Returns largest value | `max(marks)` → 95 |
| `min(list)` | Returns smallest value | `min(marks)` → 33 |

### Common List Methods

| Method | Description | Example |
|---|---|---|
| `.append(el)` | Adds element at the end | `marks.append(99)` |
| `.insert(i, el)` | Inserts element at index | `marks.insert(1, 80)` |
| `.remove(el)` | Removes first occurrence | `marks.remove(64)` |
| `.pop(i)` | Removes element at index | `marks.pop(2)` |
| `.sort()` | Sorts list in ascending order | `marks.sort()` |
| `.reverse()` | Reverses the list | `marks.reverse()` |

**Practice Question 2**

Write a program that takes names of 3 favorite foods from the user and stores them in a list. Then print the list and its length.

**Example:**

```
Input: Samosa, Pizza, IceCream
Output: ['Samosa', 'Pizza', 'IceCream']
Total items: 3
```

## 3️⃣ Tuples in Python

**Definition**

A **tuple** is a built-in data type that stores multiple values **like a list**, but it is **immutable** (cannot be changed after creation).

```
tup = (87, 64, 33, 95, 76)
print(tup[0])     # 87
```

✅ Tuples use **( )** instead of **[ ]**

**Tuple Examples**

```
t1 = ()              # Empty tuple
t2 = (1,)            # Single element tuple (comma required)
t3 = ("Samosa", "Pizza", "Burger")
```

**Immutable Nature**

```
tup = (10, 20, 30)
tup[0] = 100    # ❌ Error - cannot modify tuples
```

**Tuple Methods**

| Method | Description | Example |
|---|---|---|
| `.count(el)` | Counts occurrences of a value | `tup.count(10)` |
| `.index(el)` | Returns first index of element | `tup.index(30)` |

**Practice Question 3**

Create a tuple of your favorite 5 fruits.
Then print:

1. The total number of fruits
2. The index of one selected fruit

**Example:**

```
fruits = ("Mango", "Apple", "Banana", "Grapes", "Orange")
Output:
```

```
Total fruits: 5
Index of Banana: 2
```

## Summary

| Concept | Key Idea | Mutable | Example |
|---|---|---|---|
| **List** | Ordered, changeable sequence | Yes | `["A", "B", "C"]` |
| **Tuple** | Ordered, unchangeable sequence | No | `("A", "B", "C")` |
| **Conditionals** | Control the flow of program | — | `if`, `elif`, `else` |

## Practice Assignment

1. Ask the user for their 3 favorite movies and store them in a list.

2. Create a tuple of marks `(87, 64, 33, 95, 76)` and print the highest and lowest marks using `max()` and `min()`.

3. Write a program to check grade based on marks (A/B/C/D) using if-elif-else.